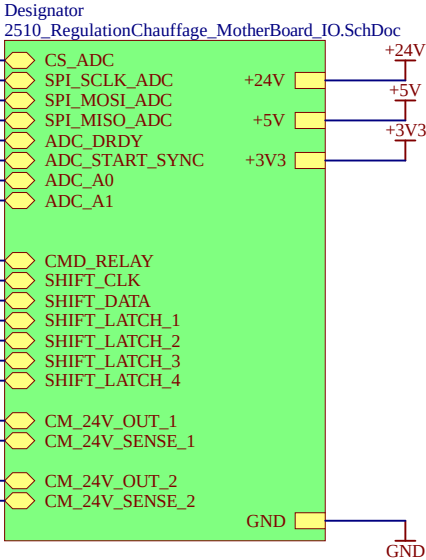
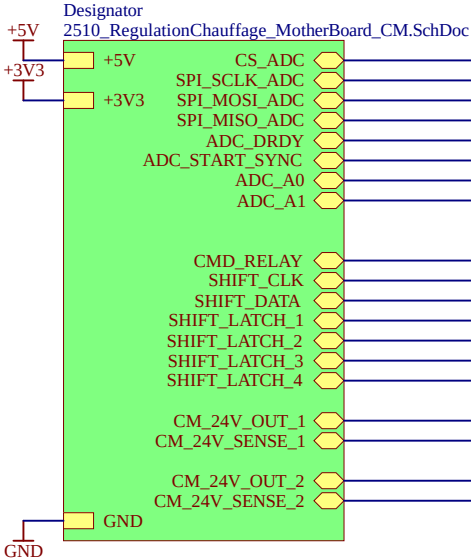
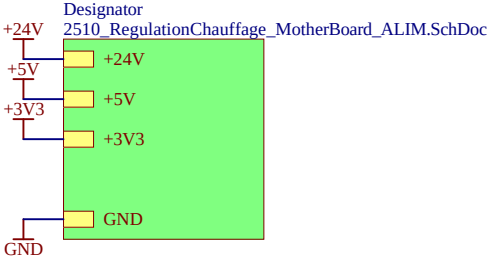
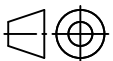
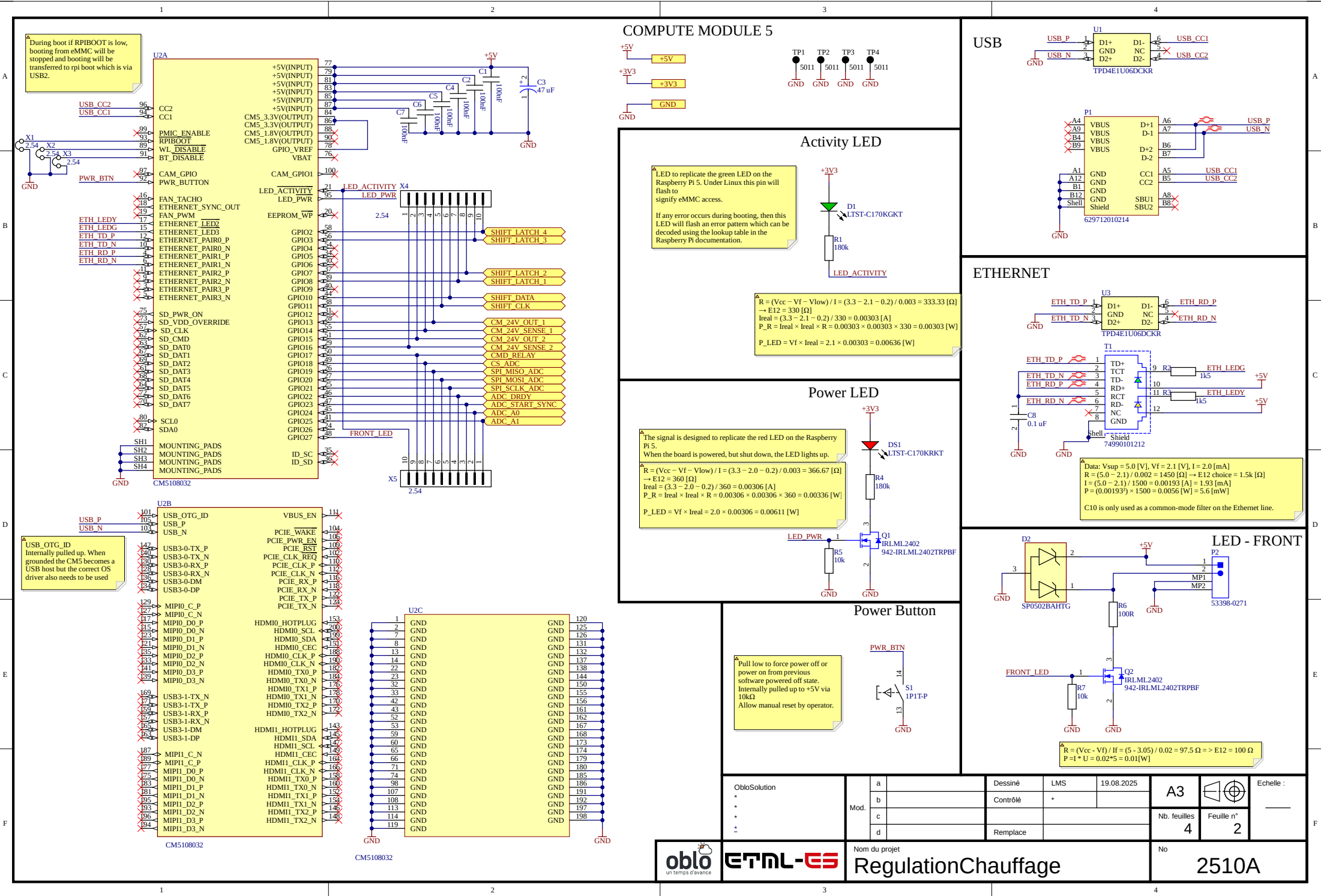
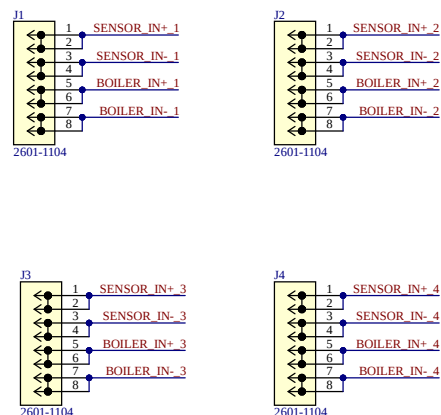




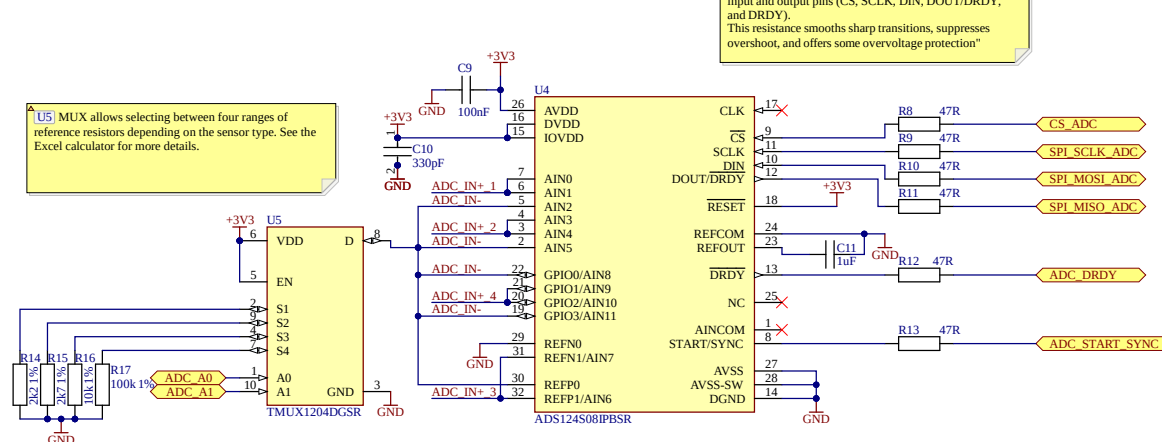
ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles	Feuille n°	
		d		Remplace			4	1	
ETML-ES		Nom du projet RegulationChauffage				 un temps d'avance		No 2510A	



I/O Sensor / Boiler



ADC



R8 to **R13** "47-Ω resistors in series with all digital input and output pins (CS, SCLK, DIN, DOUT/DRDY, and DRDY). This resistance smooths sharp transitions, suppresses overshoot, and offers some overvoltage protection"

U5 MUX allows selecting between four ranges of reference resistors depending on the sensor type. See the Excel calculator for more details.

U4 ADS124S08IPBSR – 4x 2-wire sensors

Reference: the internal 2.5 V reference is buffered and stabilized with 1 μ F between REFOUT and REFCOM.

Digital interface: the ADC communicates with the Raspberry Pi CM5 over SPI (pins CS, SCLK, MOSI, MISO). Control signals: DRDY (data ready) and START/SYNC are routed for synchronization and conversion triggering. The internal clock is used, so the CLK pin is left unconnected.

I_{AVDD}(g=4): 140 μ A typ ; 155 μ A max.
I_{DVDD}(conv): 225 μ A typ ; 300 μ A max.
I_{REFInt}: 185 μ A typ ; 280 μ A max.
I_{VBIAS} (sans charge): 10 μ A.
I_{IDAC} overhead(250 μ A): 20 μ A typ ; 35 μ A max.

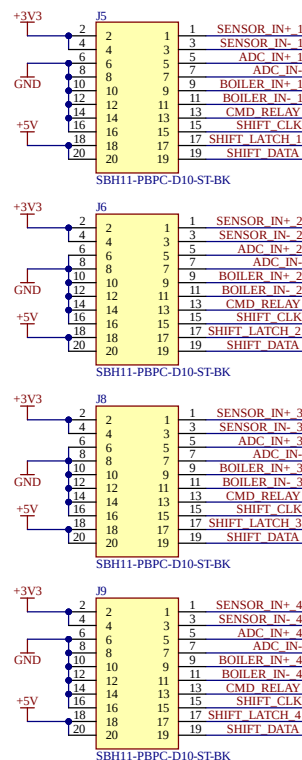
Données clé [μA] (datasheet):

I_{AVDD}(g=4,8) = 140 typ; 155 max
I_{DVDD}(conv, osc int) = 225 typ; 300 max
I_{IDAC}_overhead(250 μA) = 20 typ; 35 max
I_{IDAC} = 250

Cas VBIAS non utilisé:

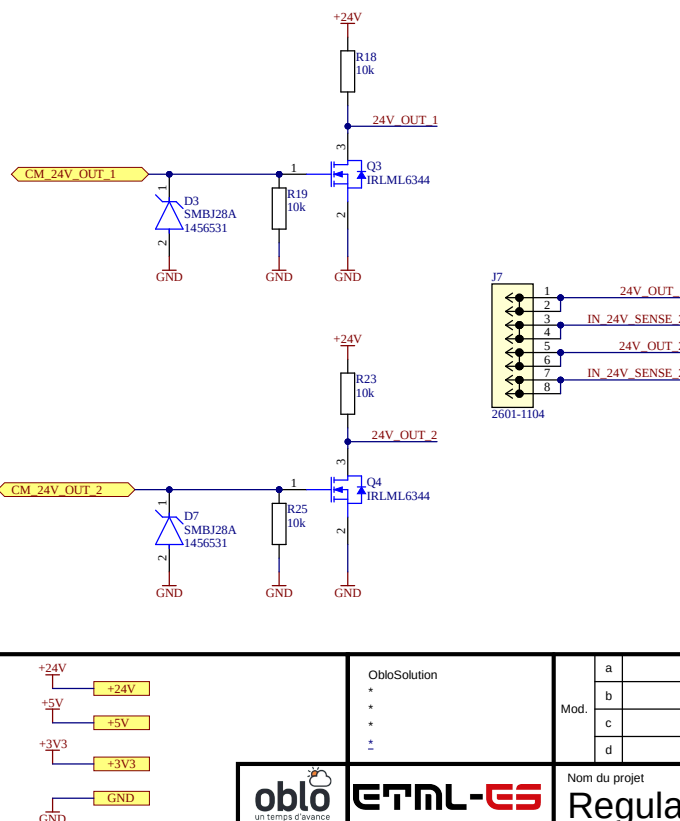
$$I_{\text{total_typ}} = 140 + 225 + 20 + 250 = 635 \text{ } [\mu\text{A}]$$

I/O DAUGHTER BOARD



CMD_RELAY	CMD_RELAY
SHIFT_CLK	SHIFT_CLK
SHIFT_DATA	SHIFT_DATA
SHIFT_LATCH 1	SHIFT_LATCH 1
SHIFT_LATCH 2	SHIFT_LATCH 2
SHIFT_LATCH 3	SHIFT_LATCH 3
	SHIFT_LATCH 4

24V - 3V3 I/O



- Output worst case (pull-up to 24 V held low through 10 k Ω)
- Input sense (24 V across divider 120 k Ω + 18 k Ω + 1 k Ω)

Output (per channel):
 $I_{out} = 24 / 10000 = 0.0024 \text{ A}$
 $P_{out} = I_{out} * I_{out} * 10000 = 0.0024 * 0.0024 * 10000 = 0.0576 \text{ W}$

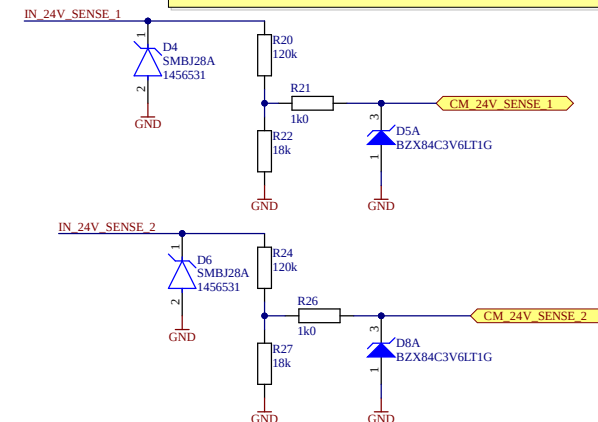
Input (per channel):
 $I_{in} = 24 / (120000 + 18000 + 1000) = 24 / 139000 = 0.000173 \text{ A}$
 $P_{in} = 24 * I_{in} = 24 * 0.000173 = 0.00415 \text{ W}$

Combined per channel (output + input):

Combined per channel (output + input):
 $I_{ch} = 0.0024 + 0.000173 = 0.002573 \text{ A}$
 $P_{ch} = 0.0576 + 0.00415 = 0.06175 \text{ W}$

Totals for 2 identical channels:

Totals for 2 identical channels:
 $I_{\text{total}} = 2 * 0.002573 = 0.005146 \text{ A}$
 $P_{\text{total}} = 2 * 0.06175 = 0.1235 \text{ W}$



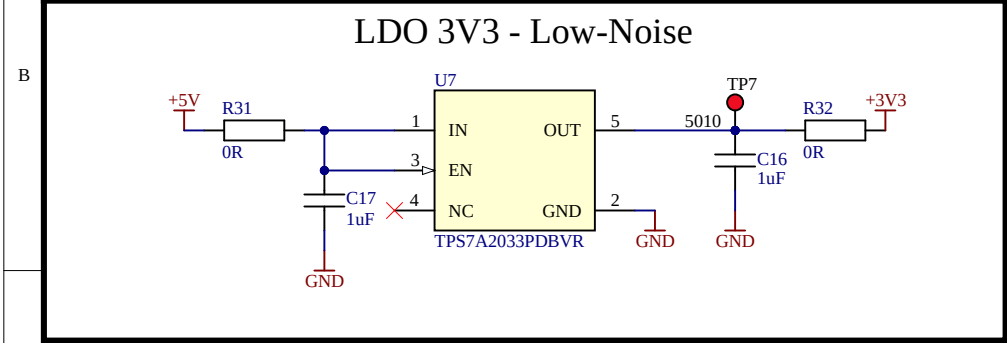
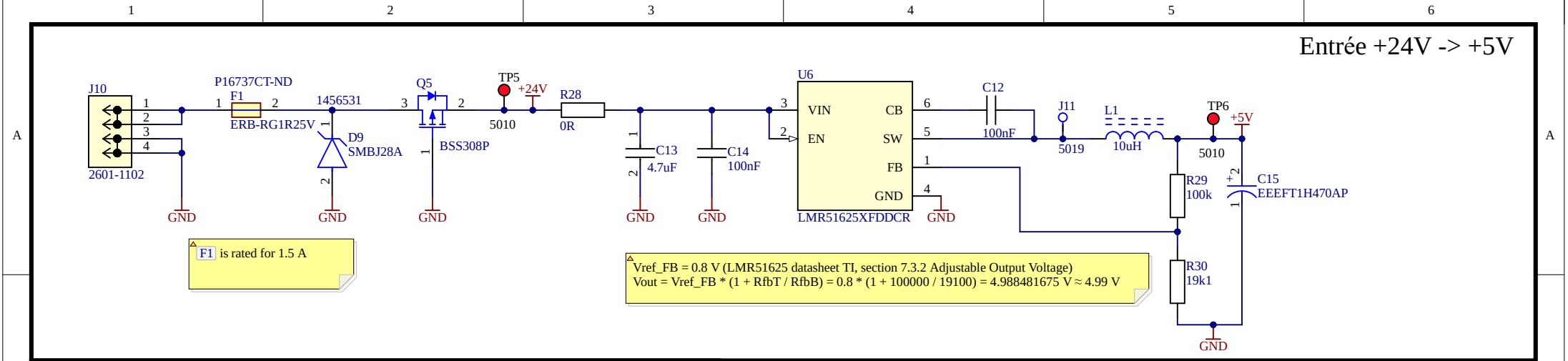
Mod.	a	Dessiné	LMS	19.08.2025	A3		Echelle : _____
	b	Contrôlé	*				
	c				Nb. feuilles 4	Feuille n° 3	
	d	Remplace					

Nom du projet	
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RegulationChauffage

	No
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2510A



D

ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles 4	Feuille n° 4	
		d		Remplace					

D

	Nom du projet RegulationChauffage		No 2510A
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Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	TargetPrice	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	5019		J11			1	0	Keystone	5019	Volume Production	DigiKey	36-5019CT-ND	0.37	0.37
	CM5108032		U2			1	0	Amphenol Communications Solutions	10164227-1004A1RLF	Volume Production	DigiKey	609-10164227-1004A1RLFCT-ND	2.55	2.55
	IRLML2402		Q1, Q2			2	0	UMW(Youtai Semiconductor Co., Ltd.)	UMWIRLML2402TR	Unknown	Digikey	4518-UMWIRLML2402TR-ND	0.43364	0.86727
	TPS7A2033PDBVR	300mA ultra-low-noise low-IQ low-dropout (LDO) linear regulator with high PSRR 5-SOT-23-40 to 125	U7			1	0	Texas Instruments	TPS7A2033PDBVR	Volume Production	Mouser	595-TPS7A2033PDBVR	0.31	0.31
	74990101212	74990101212 Telecom and Datacom Connector, 8 Contact(s), Female, Right Angle, Solder Terminal, Locking	T1			1	0	Würth Elektronik	74990101212	Unknown	Farnell	2749150	6.7	6.7
	LMR51625XFDDCR	Abaisseur de tension Régulateur à découpage IC Positif Réglable 0.8V 1 Sortie 2.5A SOT-23-6 mince, TSOT-23-6	U6			1	0	Texas Instruments	LMR51625XFDDCR	Volume Production	DigiKey	296-LMR51625XFDDCRCT-ND	2.03	2.03
	BSS308P	Canal P 30V 2 A (Ta) 500mW (Ta) Montage en surface PG-SOT23	Q5			1	0	Infineon	BSS308PEH632XTSA1	Volume Production	Farnell	2432719	0.48594	2.43
	47 uF	CAP ALUM 47UF 20% 50V SMD	C3			1	0	Panasonic	EEET11H470AP	Volume Production	Farnell	1868425	0.64428	0.64428
	EEEE11H470AP	CAP ALUM 47UF 20% 50V SMD	C15			1	0	Panasonic	EEEE11H470AP	Volume Production	Farnell	1868425	0.64428	0.64428
	0.1 uF	CAP CER 0.1 uF 10V X5R 0805	C8			1	0	Yageo Group	CC0805KRX7R9B8104	Volume Production	Farnell	3019949	0.03003	0.30028
	4.7uF	CAP CER 4.7UF 10V X5R 0805	C13			1	0	Yageo Group	CC0805KKX5R6B8475	Volume Production	DigiKey	311-1863-1-ND	0.2	0.2
	330pF	CAP CER 330pF 10V X5R 0805	C10			1	0	KEMET	C0805C333K5RACTU	Volume Production	Farnell	2521333	0.04324	0.43236
	1uF	CAPC 1uF X5R 25V 10% 2012 (0805 INCH)	C11, C16, C17	CMP-065-000041-2	Up to date	3	0	TDK	C2012K5R1E105K125AA	Volume Production	DigiKey	445-1419-1-ND	0.1866	0.55981
	100nF	CAPC 100nF X7R 50V 10% 2012 (0805 INCH)	C1, C2, C4, C5, C6, C7, C9, C12, C14	CMP-065-000038-2	Up to date	9	0	KYOCERA AVX	0805C104KAT2A	Volume Production	Farnell	499687	0.03838	0.38376
	SBH11-PBPC-D10-ST-BK	Connector Header Through Hole 20 position 0.100 (2.54mm)	J5, J6, J8, J9			4	0	Sullins	SBH11-PBPC-D10-ST-BK	Volume Production	DigiKey	S9172-ND	0.41	1.64
	2601-1102	Description	J10			1	0	WAGO	2601-1102	Unknown	Farnell	4126046	1.91	1.91
	2601-1104	Description	J1, J2, J3, J4, J7			5	0	WAGO	2601-1104	Unknown	Farnell	4126048	3.54	17.69
	ADS124S08IPBSR	IC ADC 24BIT SIGMA-DELTA 32LQFP	U4			1	0	Texas Instruments	ADS124S08IPBSR	Volume Production	DigiKey	296-45170-1-ND	11.78	11.78
	2.54	JUMPER 2.54 1x02	X1, X2, X3	CMP-054-000008-1	Up to date	3	0	Würth Elektronik	61303611121	Unknown	Mouser	710-61303611121	1.38	4.14
	LTST-C170KRKT	LED Red, 75 mW, 54 mcd, -55 to 85 degC, 2-Pin SMD, RoHS, Tape and Reel	D51			1	0	Lite-On	LTST-C170KRKT	Volume Production	DigiKey	160-1415-1-ND	0.16	0.16
	53398-0271	Male Header, Pitch 1.25 mm, 1 x 2 Position, Height 4.7 mm, -40 to 85 degC, RoHS, Tape and Reel	P2	CMP-2000-07600-1	Up to date	1	0	Molex	53398-0271	Volume Production	DigiKey	WM7606CT-ND	0.54	0.54
	ERB-RG1R25V	Micro Chip Fuse, Size [LA-W] (EIA) (mm): 3.2A-1.6 (EIA:1206), Rated Current (A): 1.25, Internal R [at 25a,f max.] (mA, f): 175, Rated Voltage (Open Circuit Voltage) [DC] (V): 63, Interrupting Rating (A): 50, Category Temp. Range (A°C): -40 to +125	F1			1	0	Panasonic	ERB-RG1R25V	Unknown	DigiKey	P16737CT-ND	0.58	0.58
	IRLML6344	MOSFET N-CH 30V 5A MICRO3/SOT23	Q3, Q4			2	0	Infineon	IRLML6344TRPBF	Volume Production	Farnell	1857299	0.39373	1.97
	TMUX1204DGSR	MULTIPLEXER	U5	CMP-04892-000013-1	Up to date	1	0	Texas Instruments	TMUX1204DGSR	Volume Production	Mouser	595-TMUX1204DGSR	0.57	0.57
	Raspberry Pi SC1570	Ordinateur monocrarte 2.4GHz 4 Cœur 8Go/0Go RAM ARM® Cortex®-A76	U2(Externe)1			1	0	Raspberry Pi	CM5008016	Unknown	Newark	20AM3761	80	80
	2.54	PIN HEADER 2.54 1x10	X4, X5	CMP-055-000013-1	Up to date	2	0	Würth Elektronik	61300411121	Unknown	Mouser	710-61300411121	0.17	0.34
	TPD4E1U06DCKR	Quad 0.8-pF, 5.5-V, $\pm$ 15-kV ESD protection diode in SOT-23 & SC70 packages 6-SC70 -40 to 125	U1, U3			2	0	Texas Instruments	TPD4E1U06DCKR	Volume Production	DigiKey	296-39347-6-ND	0.37	0.74
	0R	RES 0ohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R28, R31, R32	CMP-066-000150-2	Up to date	3	0	Vishay	CRCW0805000020EA	Volume Production	Newark	52K9721	0.022	0.066
	1k0	RES 1.0kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R21, R26	CMP-066-000187-2	Up to date	2	0	Yageo Group	RC0805FR-071KL	Volume Production	Newark	82K7536	0.004	0.008
	1k5	RES 1.5kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R2, R3	CMP-066-000189-2	Up to date	2	0	Yageo Group	RC0805FR-071K5L	Volume Production	DigiKey	311-1.50KCRCT-ND	0.01633	0.16328
	2k2 1%	RES 2.2K OHM 1% 1/8W 0805 RC0805FR-072K2L	R14			1	0	Yageo Group	RC0805FR-072K2L	Volume Production	Farnell	9237534	0.01221	0.12211
	2k7 1%	RES 2.7K OHM 1% 1/8W 0805 RC0805FR-102K7L	R15			1	0	Yageo Group	RC0805FR-102K7L	Volume Production	DigiKey	13-RC0805FR-102K7LCT-ND	0.1	0.1
	10k	RES 10kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R5, R7, R18, R19, R23, R25	CMP-066-000199-2	Up to date	6	0	Yageo Group	AC0805FR-0710KL	Volume Production	Newark	49AK0724	0.005	0.03
	18k	RES 18kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R22, R27	CMP-066-000202-2	Up to date	2	0	Yageo Group	RC0805FR-0718KL	Volume Production	Newark	49AK3089	0.006	0.012
	19k1	RES 19k1 OHM 5% 1/4W 0805	R30			1	0	Yageo Group	RC0805FR-0719K1L	Volume Production	Farnell	4061021	0.10965	0.10965
	47R	RES 47ohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R8, R9, R10, R11, R12, R13	CMP-066-000171-2	Up to date	6	0	Yageo Group	RC0805FR-0747RL	Volume Production	Newark	98K7855	0.004	0.024
	100K 1%	RES 100K OHM 1% 1/8W 0805 AC0805FR-13100KL	R17			1	0	Yageo Group	AC0805FR-13100KL	Volume Production	Farnell	4013022	0.02006	0.2006
	100k	RES 100kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R29	CMP-066-000211-2	Up to date	1	0	Yageo Group	RC0805FR-07100KL	Volume Production	Mouser	603-RC0805FR-07100KL	0.12	0.12
	100R	RES 100ohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R6	CMP-066-000175-2	Up to date	1	0	Yageo Group	RC0805FR-07100RL	Volume Production	RS	6185135	0.00454	0.00454
	120k	RES 120kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R20, R24	CMP-066-000212-2	Up to date	2	0	Yageo Group	RC0805FR-07120KL	Volume Production	Newark	49AK3045	0.004	0.008
	180k	RES 180kohm $\pm$ 1% 0.125W 2012 (0805 INCH)	R1, R4	CMP-066-000214-2	Up to date	2	0	Yageo Group	RC0805FR-07180KL	Volume Production	DigiKey	311-180KCRCT-ND	0.014	0.13995
	10k 1%	RES SMD 10K OHM 0.1% 1/8W 0805 AC0805FR-0710KL	R16			1	0	Yageo Group	AC0805FR-0710KL	Volume Production	Farnell	3495316	0.01782	0.17818
	10uH	Semi-Shielded Power Inductor, 10 uH, +/- 20%, 2.1 A, -40 to 125 degC, 2-Pin SMD, RoHS, Tape and Reel SRN3015-100M	L1			1	0	Bourns	SRN3015-100M	Volume Production	Farnell	2061634	0.40993	2.05
	1P1T-P	SW 600X600X700 24V 50mA THT	S1	CMP-058-000006-2	Up to date	1	0	TE Connectivity	1-1825910-4	Volume Production	Newark	95M4263	0.019	0.019
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2, TP3, TP4			4	0	Keystone	5011	Volume Production	Newark	88H2535	0.137	0.548
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP5, TP6, TP7			3	0	Keystone	5010	Volume Production	Mouser	534-5010	0.37	1.11
	SP0502BAHTG	TVS DIODE 5.5V 8.5V SOT23-3	D2			1	0	Littelfuse	SP0502BAHTG	Volume Production	Farnell	1827631	0.69775	0.69775
	SMBJ28A	TVS DIODE 28V 45.4V DO214AA	D3, D4, D6, D7, D9			5	0	Littelfuse	SMBJ28A	Volume Production	Farnell	1456531	0.25169	1.26
	LTST-C170KGKT	Ultra Bright AlInGaP Chip LED, 75 mW, -55 to 85 degC, 2-Pin SMD (0805), RoHS, Tape and Reel	D1			1	0	Lite-On	LTST-C170KGKT	Volume Production	DigiKey	160-1414-1-ND	0.18	0.18
	629712010214	USB 2.0 Type C Plug Vertical SMT	P1			1	0	Würth Elektronik	629712010214	Unknown	DigiKey	732-629712010214CT-ND	4.55	4.55
	BZX84C3V6LT1G	Zener Voltage Regulator, 225 mW, 3-Pin SOT-23, Pb-Free, Tape and Reel	D5, D8	CMP-1060-01139-1	Up to date	2	0	onsemi	BZX84C3V6LT1G	Volume Production	Farnell	2317468	0.11339	0.56694

BOM Cost		PROJET:	2510_RegulationChauffage_Motherboard.PrjPcb									ETML-ES Rue de Sébeillon 12 1004 Lausanne		
Date:	29.08.2025	Variante:	None								Quantité de production (nb PCBs) : 1		duction (nb PCBs) :	100
Par:	LMS	Source:	2510_RegulationChauffage_Motherboard.BomDoc								Devise CHF		Devise : CHF	
Line #	Designator	Value	Name	Manufacturer 1	Manufacturer Part Number 1	Quantit y	Supplier 1	Supplier Part Number 1	Supplier Order Qty 1	Supplier Unit Price 1	Supplier Subtotal 1	Supplier Order Qty 1	Supplier Unit Price 1	Supplier Subtotal 1
	C1, C2, C4, C5, C6, C7, C9, C12, C14	100nF	100nF	KYOCERA AVX	08055C104KAT2A	9	Farnell	499687	9	0.0308	0.2772	900	0.0308	27.72
	C3		47 uF	Panasonic	EEEF1H470AP	1	Farnell	1868425	1	0.415	0.415	100	0.415	41.5
	C8	0.1 uF	0.1 uF	Yageo Group	CC0805KRX7R9BB104	1	Farnell	3019949	1	0.0241	0.0241	100	0.0241	2.41
	C10	330pF	330pF	KEMET	C0805C331K5RACTU	1	Farnell	2521333	1	0.0347	0.0347	100	0.0347	3.47
	C11, C16, C17	1µF	1uF	TDK	C2012X5R1E105K125AA	3	Mouser	810-C2012X5R1E105K-3	3	0.061	0.183	300	0.061	18.3
	C13	4.7 uF	4.7uF	Yageo Group	CC0805KKX5R6BB475	1	DigiKey	311-1863-1-ND	1	0.0923	0.0923	100	0.0923	9.23
	C15		EEEF1H470AP	Panasonic	EEEF1H470AP	1	Farnell	1868425	1	0.415	0.415	100	0.415	41.5
	D1		LTST-C170KGKT	Lite-On	LTST-C170KGKT	1	DigiKey	160-1414-1-ND	1	0.14446	0.14446	100	0.14446	14.446
	D2		SP0502BAHTG	Littelfuse	SP0502BAHTG	1	Farnell	1827631	1	0.56	0.56	100	0.56	56
	D3, D4, D6, D7, D9		SMBJ28A	Littelfuse	SMBJ28A	5	Farnell	1456531	5	0.202	1.01	500	0.202	101
	D5, D8		BZX84C3V6LT1G	onsemi	BZX84C3V6LT1G	2	Farnell	2317468	2	0.091	0.182	200	0.091	18.2
	DS1		LTST-C170KRKT	Lite-On	LTST-C170KRKT	1	DigiKey	160-1415-1-ND	1	0.12841	0.12841	100	0.12841	12.841
	F1		ERB-RG1R25V	Panasonic	ERB-RG1R25V	1	DigiKey	P16737CT-ND	1	0.38203	0.38203	100	0.38203	38.203
	J1, J2, J3, J4, J7		2601-1104	WAGO	2601-1104	5	Farnell	4126048	5	2.7	13.5	500	2.7	1350
	J5, J6, J8, J9		SBH11-PBPC-D10-ST-BK	Sullins	SBH11-PBPC-D10-ST-BK	4	DigiKey	S9172-ND	4	0.27849	1.11396	400	0.27849	111.396
	J10		2601-1102	WAGO	2601-1102	1	Farnell	4126046	1	1.53	1.53	100	1.53	153
	J11		5019	Keystone	5019	1	DigiKey	36-5019CT-ND	1	0.29695	0.29695	100	0.29695	29.695
	L1		10uH	Bourns	SRN3015-100M	1	Farnell	2061634	1	0.329	0.329	100	0.329	32.9
	P1		629712010214	Würth Elektronik	629712010214	1	DigiKey	732-629712010214CT-ND	1	3.65	3.65	100	3.29	329
	P2		53398-0271	Molex	53398-0271	1	DigiKey	WM7606CT-ND	1	0.43339	0.43339	100	0.43339	43.339
	Q1, Q2		IRLML2402	UMW(Youtai Semiconductor Co.,	UMWIRLML2402TR	2	Digikey	4518-UMWIRLML2402TR-ND	2	0.34803	0.69606	200	0.34803	69.606
	Q3, Q4		IRLML6344	Infineon	IRLML6344TRPBF	2	Farnell	1857299	2	0.316	0.632	200	0.316	63.2
	Q5		BSS308P	Infineon	BSS308PEH6327XTSA1	1	Farnell	2432719	1	0.39	0.39	100	0.39	39
	R1, R4	180k	180k	Yageo Group	RC0805FR-07180KL	2	DigiKey	311-180KCRCT-ND	2	0.01123	0.02246	200	0.01123	2.246
	R2, R3	1.5k	1k5	Yageo Group	RC0805FR-071K5L	2	DigiKey	311-1.50KCRCT-ND	2	0.0131	0.0262	200	0.0131	2.62
	R5, R7, R18, R19, R23, R25	10k	10k	Yageo Group	AC0805FR-0710KL	6	Newark	49AK0724	6	0.00401	0.02406	600	0.00401	2.406
	R6	100	100R	Yageo Group	RC0805FR-07100RL	1	RS	6185135	1	0.00364	0.00364	100	0.00364	0.364
	R8, R9, R10, R11, R12, R13	47	47R	Yageo Group	RC0805FR-0747RL	6	Newark	98K7855	6	0.01284	0.07704	600	0.01284	7.704
	R14		2k2 1%	Yageo Group	RC0805FR-072K2L	1	Farnell	9237534	1	0.0098	0.0098	100	0.0098	0.98
	R15		2k7 1%	Yageo Group	RC0805FR-102K7L	1	DigiKey	13-RC0805FR-102K7LCT-ND	1	0.01284	0.01284	100	0.01284	1.284
	R16		10k 1%	Yageo Group	AC0805FR-0710KL	1	Farnell	3495316	1	0.0143	0.0143	100	0.0143	1.43
	R17		100k 1%	Yageo Group	AC0805FR-13100KL	1	Farnell	4013022	1	0.0161	0.0161	100	0.0161	1.61
	R20, R24	120k	120k	Yageo Group	RC0805FR-07120KL	2	Newark	49AK3045	2	0.00321	0.00642	200	0.00321	0.642
	R21, R26	1k	1k0	Yageo Group	RC0805FR-071KL	2	Newark	82K7536	2	0.00321	0.00642	200	0.00321	0.642
	R22, R27	18k	18k	Yageo Group	RC0805FR-0718KL	2	Newark	49AK3089	2	0.00482	0.00964	200	0.00482	0.964
	R28, R31, R32	0	0R	Vishay	CRCW08050000Z0EA	3	Mouser	71-CRCW0805-0-E3	3	0.01364	0.04092	300	0.01364	4.092
	R29	100k	100k	Yageo Group	RC0805FR-07100KL	1	Newark	05M4536	1	0.01124	0.01124	100	0.01124	1.124
	R30		19k1	Yageo Group	RC0805FR-0719K1L	1	Farnell	4061021	1	0.01	0.01	100	0.01	1
	S1	1P1T-P	1P1T-P	TE Connectivity	1-1825910-4	1	Newark	95M4263	1	0.01525	0.01525	100	0.01525	1.525
	T1		74990101212	Würth Elektronik	74990101212	1	Farnell	2749150	1	5.38	5.38	100	4.11	411
	TP1, TP2, TP3, TP4		5011	Keystone	5011	4	Newark	88H2535	4	0.10995	0.4398	400	0.10995	43.98
	TP5, TP6, TP7		5010	Keystone	5010	3	Mouser	534-5010	3	0.23515	0.70545	300	0.23515	70.545
	U1, U3		TPD4E1U06DCKR	Texas Instruments	TPD4E1U06DCKR	2	DigiKey	296-39347-6-ND	2	0.19904	0.39808	200	0.19904	39.808
	U2		CM5108032	Amphenol Communications	10164227-1004A1RLF	1	DigiKey	609-10164227-1004A1RLFCT-ND	1	2.05	2.05	100	2.05	205
	U2(Externe)1		Raspberry Pi SC1570	Raspberry Pi	CM5008016	1	Newark	20AM3761	1	64.21	64.21	100	64.21	6421
	U4		ADS124S08IPBSR	Texas Instruments	ADS124S08IPBSR	1	DigiKey	296-45170-1-ND	1	9.45	9.45	100	6.3242	632.42
	U5		TMUX1204DGSR	Texas Instruments	TMUX1204DGSR	1	Mouser	595-TMUX1204DGSR	1	0.45747	0.45747	100	0.45747	45.747
	U6		LMR51625XFDDCR	Texas Instruments	LMR51625XFDDCR	1	DigiKey	296-LMR51625XFDDCRCT-ND	1	1.63	1.63	100	1.63	163
	U7		TPS7A2033PDBVR	Texas Instruments	TPS7A2033PDBVR	1	Mouser	595-TPS7A2033PDBVR	1	0.2488	0.2488	100	0.2488	24.88
	X1, X2, X3	2.54	2.54	Würth Elektronik	61303611121	3	Farnell	2356174	3	0.65427	1.96281	300	0.65427	196.281
	X4, X5	2.54	2.54	Würth Elektronik	61300411121	2	DigiKey	732-5317-ND	2	0.11794	0.23588	200	0.11794	23.588
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